MINI PROJECT REPORT

On

“EDUVAULT: DIGITAL LOCKER AND ACADEMIC CREDENTIAL VERIFICATION PLATFORM”

By

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Under The Guidance Of

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Submitted To

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CERTIFICATE

This is to certify that, **Akshay Darji** (PRN: **1272250311**) and **Priyesh Nandanwar** (PRN: **1272250279**), students of **SY MCA, Semester III** have completed Mini Project entitled **"EduVault: Digital Locker and Academic Credential Verification Platform"** under guidance of **Mrs. Sukanya Akolkar** in partial fulfilment of requirement for the award of **Master of Computer Applications** prescribed under **Dr. Vishwanath Karad MIT World Peace University, Pune**.

This project is the record of authentic work carried out by them during the academic year **2026-2027**.

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DECLARATION

We, **Akshay Darji** and **Priyesh Nandanwar**, on behalf of our team hereby declare that this Mini Project report entitled **"EduVault: Digital Locker and Academic Credential Verification Platform"** is an authentic record of our own work carried out under the supervision of **Mrs. Sukanya Akolkar**.

We further declare that the content of this report, including all design, coding, testing, and analysis, is original and has not been submitted previously for any other degree or diploma at this or any other institution.

We take sole and full responsibility for any errors, omissions, or misrepresentations found within this report.

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1. INTRODUCTION

Higher education institutions generate immense volumes of academic records, including degree certificates, semester marksheets, co-curricular awards, industrial internship validations, and technical project portfolios. However, the management, storage, verification, and dissemination of these records have historically remained fragmented across manual paper archives, disparate department files, and unverified digital scans. This fragmentation causes severe delays in institutional background checks, fosters academic fraud, burdens placement departments with manual spreadsheet compilation, and isolates graduated alumni from active university mentorship networks.

This chapter describes existing credential management practices and their limitations, introduces the proposed **EduVault** ecosystem, defines the project scope, outlines hardware and software operational environments, and justifies the technology selections powering the platform.

1.1 EXISTING SYSTEM, PROPOSED SYSTEM AND NEED AND OBJECTIVE OF SYSTEM

1.1.1 Existing System

Academic credential management and campus placement coordination currently depend on four loosely coupled, manual procedures:

(a) Physical Paper Transcripts & Unauthenticated PDF Scans.
Students submit physical photocopies or generic PDF scans of their credentials during semester evaluations and campus placement registrations. These files are not cryptographically tied to an institutional source of truth. Consequently, they are vulnerable to alteration, forged seals, and manipulated marks using basic graphic editing tools.

(b) Manual Administrative Verification Workflows.
Institutions establish academic verification committees to authenticate student credentials. However, verifiers must manually retrieve physical ledger books or query disconnected database spreadsheets. This creates heavy administrative backlogs, with third-party background verification agencies frequently waiting two to four weeks for manual transcript confirmations.

(c) Disconnected Placement & Talent Sourcing.
Training & Placement Officers (TPO) spend countless hours compiling candidate data (CGPA, active backlogs, departmental affiliations, and project links) through Google Forms or spreadsheets. This self-reported data is prone to human error, lacks instant verification, and cannot be dynamically presented to corporate recruiters.

(d) Static Alumni Communication Silos.
Upon graduation, alumni contact records deteriorate into static contact sheets. Universities lack an interactive, database-backed registry where alumni can advertise current company designations, share locations, and declare availability for student mentorship.

1.1.2 Proposed System

EduVault is an end-to-end, enterprise-grade Digital Locker, Academic Credential Verification, and Career Networking Platform. It unites students, verification committees, placement officers, alumni, and external recruiters within a secure, role-governed web application. The platform operates across six core functional tiers:

1. **Authentication & Role Governance Tier:** Issues signed JSON Web Tokens (JWT) using Django REST Framework SimpleJWT, routing users automatically into one of four tailored portals: `STUDENT`, `ALUMNI`, `COMMITTEE`, or `PLACEMENT`.
2. **Student Digital Vault:** Enables students to upload, categorize, and track semester marksheets, degree certificates, and achievement files (≤ 5 MB). Uploaded records initiate in a `PENDING` state.
3. **Academic Committee Verification Pipeline:** Provides faculty reviewers with a real-time review queue, in-app full-screen document previewers, instant verification approvals, or rejection feedback mechanisms.
4. **Cryptographic Interview Key & QR Engine:** Generates unique UUID-based cryptographic endpoints (e.g. `/student/:id` and `/verify/:id`) that render as high-resolution QR codes, enabling corporate recruiters to scan and inspect verified student credentials on mobile devices instantly.
5. **Placement & Corporate Relations Dashboard:** Empowers placement teams with aggregate talent metrics (Total Students, Verified Ratio, Average CGPA), instant text search, and multi-tier CGPA threshold pills (≥ 9.0, ≥ 8.0, ≥ 7.5, ≥ 7.0).
6. **Dynamic Alumni Network:** An interactive, database-backed directory allowing graduated alumni to publish current roles, employers, geographic locations, and mentorship availability tags.

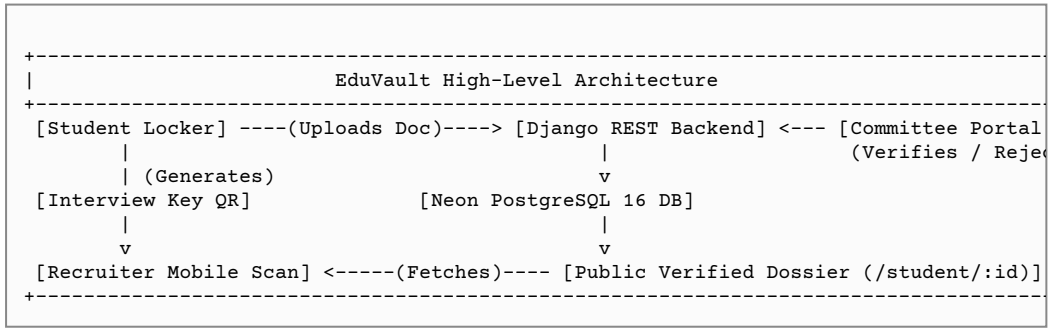


Figure 1.1: High-level architectural dataflow of the EduVault platform.

1.1.3 Comparative Analysis

Table 1.1 compares traditional institutional document management with EduVault across eight operational parameters.

Table 1.1: Comparison of existing approaches with EduVault across operational parameters.

Sr.	Parameter	Manual Paper / Files	Generic Cloud Drive	EduVault Platform
1	**Verification Mechanism**	Manual stamp, physical ledger lookup	Unverified link sharing; no institutional approval	Formal academic committee pipeline with immutable verification timestamps
2	**Fraud Resistance**	Low; easily forged with photocopies	Zero; raw uploaded PDFs cannot be trusted	High; UUID tokens and institutional QR verification ensure authenticity
3	**Recruiter Inspection**	Requires physical documents at interview	Requires manual downloading & unorganized viewing	1-click QR Code scan directly to mobile-ready verified dossier
4	**Role Governance**	None; unorganized file cabinets	Basic link permissions (Viewer/Editor)	Strict 4-Role RBAC: Student, Alumni, Committee, Placement Team
5	**Placement Analytics**	Manual spreadsheet tallying	Disconnected sheets prone to student typos	Live talent pool metrics, dynamic CGPA threshold filters, and verified pills
6	**Alumni Integration**	Static address books	None; personal LinkedIn networks only	Dynamic database registry with mentorship filters and company badges
7	**Document Lifecycle**	Uncontrolled paper lifecycle	Unmonitored file edits	State machine: PENDING → VERIFIED or REJECTED with feedback
8	**Hosting Cost**	High paper & courier costs	Paid workspace licenses	100% Free Production Deployment (Vercel + Render + Neon PostgreSQL)

1.1.4 Need for the System

Eliminating Academic Credential Fraud: With increasing competition in campus placements, recruiters face fabricated resumes and altered CGPA reports. EduVault eliminates fraud by anchoring credentials to institutional committee verification.

Instant Verification Latency Reduction: Employers spend up to two weeks on third-party verification. Scanning an EduVault Interview Key reduces verification latency to under 3 seconds.

Institutional Digital Transformation: Aligns the university with paperless, sustainable e-governance standards while providing students with a permanent digital portfolio.

1.1.5 Objectives of the System

The engineering objectives are stated so that each can be verified by inspection, test, or measurement during the evaluation phase.

Table 1.2: Engineering objectives and verification criteria.

ID	Objective	Verification Criterion
O1	Implement a decoupled full-stack architecture with Django REST Framework backend and React 18 frontend.	Frontend and backend communicate purely over JSON REST APIs with no server-side HTML rendering.
O2	Establish Role-Based Access Control (RBAC) across four user classes: Student, Alumni, Committee, and Placement Team.	Role permissions are strictly enforced on backend endpoints (HTTP 403 Forbidden on unauthorized requests) and frontend route guards.
O3	Develop a Digital Locker allowing document uploads with strict size and extension validation.	Uploads ≤ 5 MB in PDF, JPG, and PNG succeed; oversized or executable files are rejected with HTTP 400.
O4	Build an Academic Committee review pipeline with approve and reject actions.	Committee reviews update document state to `VERIFIED` (with timestamp) or `REJECTED` with mandatory student feedback.
O5	Generate dynamic cryptographic QR Codes for physical and digital interview verification.	Mobile camera scan of the generated QR code opens the public dossier (`/student/:id`) with verified records on mobile browsers.
O6	Create a Talent Discovery Portal for Placement Officers with multi-criteria filtering.	Placement dashboard renders dynamic talent pool metrics and correctly filters candidates by CGPA thresholds (e.g. ≥ 8.0).
O7	Deploy the full-stack system to free production cloud tiers without hosting expenditures.	Frontend is live on Vercel Edge CDN, backend on Render WSGI service, and database on Neon Serverless PostgreSQL.
O8	Provide authenticated media file serving in production environments.	Uploaded certificates open cleanly from the live Render backend media endpoint with zero 404 routing errors.

1.2 SCOPE OF WORK

1.2.1 Functional Scope

The functional scope encompasses all user-facing interactions across the platform:

- **User Registration & Profiling:** Account creation with department, college, PRN, phone, and role assignment.
- **JWT Session Security:** Token issuance, automated refresh token rotation via Axios interceptors, and logout handling.
- **Student Vault Management:** Uploading, previewing, and deleting unverified academic documents.
- **Institutional Committee Queue:** Queue sorting, filtering by status/category, full-screen document inspection, single-click verification, and rejection with feedback.
- **Recruitment & Placement Engine:** Live talent metrics, candidate search, CGPA threshold filtering, and verified dossier export.
- **Dynamic Alumni Network:** Real-time alumni registry with employer, role, location, and mentorship availability filters.
- **Public QR Verification:** Tokenized, unauthenticated read-only candidate dossier view for external corporate recruiters.

1.2.2 Technical Scope

- **Backend REST Service:** Python 3.11 with Django 6.0 and Django REST Framework, served via Gunicorn WSGI.
- **Persistence:** PostgreSQL 16 (Neon Serverless Cloud) managed through Django ORM with ACID-compliant transactions and foreign key cascades.
- **Frontend Application:** React 18 with Vite build engine, React Router DOM v6, Tailwind CSS, Lucide React icons, and QRCode.react.
- **Cloud Infrastructure:** Vercel for SPA hosting with edge routing; Render for Python WSGI container execution.
- **CORS & Reverse Proxy:** Reverse proxy forwarding in Vite for mobile LAN inspection, and open cross-origin production bindings.

1.2.3 Out of Scope and Boundaries

The following are explicitly excluded from the present work:

1. **Dynamic Analysis & OCR:** The system stores and displays files; automatic character recognition (OCR) of scanned certificates is excluded from the current release.
2. **Automated Grading:** The platform tracks CGPA and marksheets; it does not compute semester examination grades from scratch.
3. **Payment Gateway Integration:** Administrative transcript fees are handled externally; online financial transactions are out of scope.
4. **Native Mobile Apps:** The system is built as a fully responsive Web App; native compilation for Android APK or iOS IPA is not implemented.

1.3 OPERATING ENVIRONMENT – HARDWARE AND SOFTWARE

1.3.1 Hardware Requirements

Table 1.3: Hardware requirements by deployment role.

Role	Component	Minimum Specification	Recommended Specification
Development Workstation	Processor	Dual-Core 64-bit CPU, 2.0 GHz	Intel Core i5 / AMD Ryzen 5, 2.8 GHz or higher
	RAM	8 GB DDR4	16 GB DDR4/DDR5
	Storage	256 GB SSD (500 MB free for project)	512 GB NVMe SSD
	Network	10 Mbps Broadband	50 Mbps Wi-Fi / Ethernet
Cloud Production Server	Compute	0.5 vCPU (Render Free Tier)	1.0 vCPU Dedicated Container
	RAM	512 MB RAM	1 GB RAM or higher
	Database Storage	0.5 GB Serverless PostgreSQL (Neon)	5 GB Persistent Cloud PostgreSQL
	Bandwidth	100 Mbps Cloud Ingress/Egress	1 Gbps Cloud Pipe with CDN Caching
Client Access Device	Desktop / Mobile	Dual-Core 1.6 GHz / Android 8.0+	Modern Smartphone / Laptop with Camera
	Client RAM	2 GB (Mobile) / 4 GB (Desktop)	6 GB+ Mobile / 8 GB Desktop
	Network	3G / 4G Mobile Data (2 Mbps)	4G / 5G / Broadband Wi-Fi (10 Mbps+)

1.3.2 Software Requirements

Table 1.4: Software requirements and tooling versions.

Category	Software / Library	Version	Purpose in EduVault
Operating System	Windows 11 / Ubuntu 22.04 LTS	64-bit	Development and cloud container host runtime
Language Runtime	Python	3.11.x	Backend server runtime and business logic
Backend Framework	Django & Django REST Framework	Django 6.0 / DRF 3.16	RESTful API routing, serialization, and ORM
Authentication	SimpleJWT (rest_framework_simplejwt)	5.3.x	Stateless JSON Web Token auth & refresh handling
Frontend Runtime	Node.js & npm	Node v20.x / npm 10.x	Client build environment and package manager
Frontend Framework	React & Vite	React 18.x / Vite 8.x	Component-driven single page application (SPA)
Styling	Tailwind CSS	3.4.x	Utility-first responsive design and micro-interactions
Icons & QR	Lucide React & QRCode.react	Latest	Vector UI iconography and SVG QR code rendering
Database	PostgreSQL	16.x	ACID-compliant relational database persistence
Static Asset Server	WhiteNoise	6.x	Gzip-compressed static file serving for WSGI
Production WSGI	Gunicorn	25.x	Production WSGI HTTP server on Render

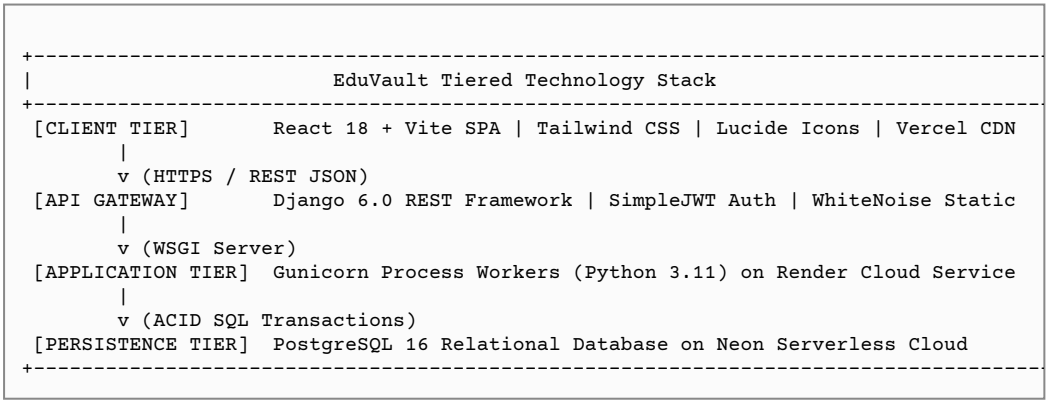


Figure 1.2: Four-tier technology architecture of EduVault.

1.4 DESCRIPTION OF TECHNOLOGY USED

1.4.1 Python 3.11 Runtime Environment

Python 3.11 is used as the primary backend programming language. It delivers substantial execution speed improvements, optimized memory utilization, and precise error tracebacks. Its rich package ecosystem simplifies database communication, file validation, and token cryptography.

1.4.2 Django 6.0 and Django REST Framework (DRF)

Django serves as the backend application framework, offering built-in security mechanisms against Cross-Site Request Forgery (CSRF), SQL Injection, and Cross-Site Scripting (XSS). Django REST Framework provides high-speed serialization, multipart form parsing for uploads, and role permission validation.

1.4.3 SimpleJWT Token Authentication Architecture

JSON Web Tokens (RFC 7519) provide stateless session handling. Upon valid authentication, the server signs an access token (5-minute lifetime) and refresh token (24-hour lifetime). Tokens are transmitted via standard HTTP `Authorization: Bearer <token>` headers.

1.4.4 React 18 and Vite Tooling Engine

React 18 provides a component-driven declarative UI. Reactive state hooks (`useState`, `useEffect`, `useMemo`, `useRef`) maintain responsive views without full-page reloads. Vite provides instant development server startup and optimized Rollup production bundling.

1.4.5 Tailwind CSS and Lucide React Visual Components

Tailwind CSS provides a utility-first styling system, enforcing design token consistency across padding, typography, and borders without bulky style files. Lucide React provides intuitive vector SVG icons across all portals.

1.4.6 PostgreSQL 16 and Neon Serverless Persistence

PostgreSQL 16 guarantees relational data integrity. Hosted on Neon Serverless PostgreSQL, it provides auto-scaling compute, scale-to-zero capabilities, and strict ACID transaction safety.

1.4.7 QRCode.react and Cryptographic Token Verification

Client-side QR generation is powered by `qrcode.react`. Each QR code encodes an authenticated public URL (e.g. `https://eduvaul.vercel.app/student/:id`), allowing recruiters to scan credentials directly from physical mobile cameras.

1.4.8 Gunicorn WSGI and WhiteNoise Production Static Server

Gunicorn runs as the production Python WSGI HTTP server, handling concurrent client worker processes. WhiteNoise integrates directly into Django's middleware pipeline, serving collected static assets with brotli/gzip compression and immutable caching headers.

```
+-------------------------------------------------------------------------------------+
|                          EduVault Technology Ecosystem                              |
+-------------------------------------------------------------------------------------+
| Layer              | Selected Tooling           | Key Responsibility                |
|--------------------+----------------------------+-----------------------------------
| Presentation       | React 18 + Vite            | Reactive UI and SPA routing
| Styling            | Tailwind CSS 3.4           | Utility-first responsive design
| API Backend        | Django 6.0 + DRF 3.16      | REST API endpoints and validation
| Authentication     | SimpleJWT                  | Signed token authorization
| Persistence        | PostgreSQL 16 (Neon)       | Relational ACID database
| Asset Delivery     | WhiteNoise 6.12            | Static assets via WSGI pipeline
| Production Server  | Gunicorn 25.1              | Multi-worker WSGI HTTP server
| Cloud Hosting      | Vercel (FE) + Render (BE)  | Edge CDN and Linux cloud container
+-------------------------------------------------------------------------------------+
```

Figure 1.3: Summary breakdown of the EduVault technology ecosystem.

2. PROPOSED SYSTEM

EduVault is architected as a decoupled, multi-portal academic credential and placement platform. Each tier maintains a distinct responsibility and communicates with neighboring tiers purely through defined REST API contracts.

The design rests on four core principles:

- **Decoupling:** The React client knows only the REST API contract. The Django backend knows nothing about browser DOM rendering. Either can be modified independently.
- **Role Governance:** Complete segregation of duties across Students, Alumni, Committee Members, and Placement Teams. Non-student features are inaccessible from student views.
- **Immediate Verifiability:** Every verified document carries an immutable timestamp and cryptographic UUID, instantly resolvable via physical QR codes.
- **Zero-Cost Cloud Deployment:** Architected specifically to operate on high-availability, free production cloud tiers without institutional capital investment.

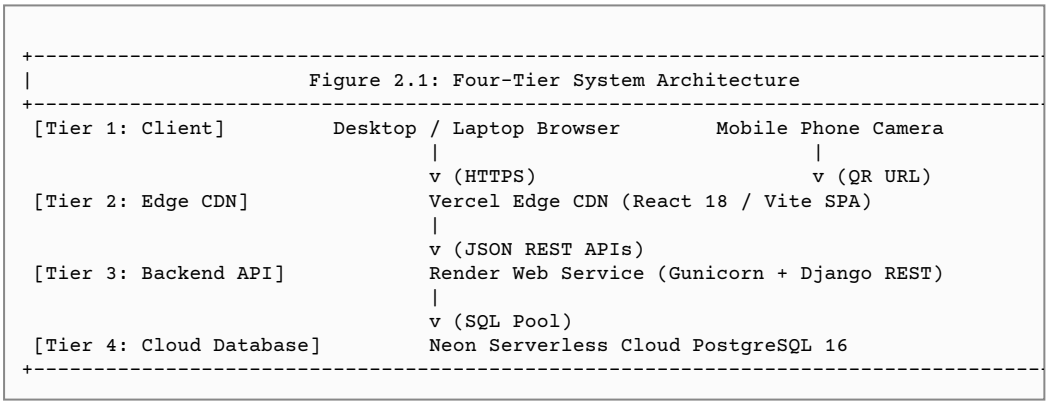


Figure 2.1: Four-tier system architecture of EduVault.

2.1 FEASIBILITY STUDY

2.1.1 Technical Feasibility

Architecture Stability: Proven decoupled stack using Python, Django, React, and PostgreSQL with extensive documentation.
Database Integrity: PostgreSQL 16 guarantees ACID properties and cascades foreign key deletions cleanly.
Security & Cryptography: Passwords hashed using PBKDF2 with SHA-256 salts; document tokens use UUIDv4 cryptographic strings.
Cloud Scalability: Vercel provides instant global caching; Render manages container execution seamlessly.

2.1.2 Operational Feasibility

Zero Client Installation: Users access the platform via standard browsers on desktops or mobile phones without installing local plugins.
Administrative Efficiency: Decreases document verification turnaround from 2–3 weeks to under 60 seconds per credential.
Recruiter Usability: Recruiters scan candidate QR codes to view verified credentials without logging in.

2.1.3 Economic Feasibility

Open-Source Software: Zero software licensing fees.
Free Production Cloud Tiers: Deployed on Vercel (Frontend), Render (Backend), and Neon (PostgreSQL), incurring zero hosting fees.
Institutional Savings: Eliminates paper transcript printing, courier fees, and manual audit labor.

2.1.4 Feasibility Conclusion

Table 2.3: Overall feasibility conclusion.

Dimension	Verdict	Principal Evidence
Technical	Feasible	Mature tech stack, REST contract, ACID PostgreSQL, sub-second client rendering.
Operational	Feasible	Browser-based, zero setup for end users, instant mobile QR verification.
Economic	Feasible	Zero licensing fees, hosted on 100% free production cloud tiers.
Security & Privacy	Feasible	PBKDF2 hashing, JWT tokens, RBAC permission guards, and file validation.

2.2 USER REQUIREMENTS

2.2.1 User Classes and Actor Profiles

Table 2.4: EduVault user classes and actor profiles.

User Class	Primary Responsibilities	Key Interactions	Authentication Method
Student	Upload academic marksheets, degree certificates, project portfolios, and internship records.	Digital Locker, Certificates Vault, Project Showcase, Interview Key QR Modal.	Email & Password, JWT (Role: STUDENT)
Alumni	Publish current industry employment, designation, location, and mentor availability.	Alumni Network Directory, Profile Manager, Student Mentorship Connection.	Email & Password, JWT (Role: ALUMNI)
Academic Committee	Review student credential submissions, inspect files, verify or reject with feedback.	Verification Queue, Document Inspector Modal, Rejection Feedback Dispatcher.	Email & Password, JWT (Role: COMMITTEE)
Placement Team	Discover job-ready candidates, filter talent by CGPA thresholds, and inspect dossiers.	Placement Dashboard, CGPA Filter Pills, Candidate Drawer, Verification Status Filter.	Email & Password, JWT (Role: PLACEMENT)
Public Recruiter	Scan student QR codes during campus interviews to verify authentic credentials.	Public Candidate Dossier (`/student/:id`), Certificate Viewer.	Unauthenticated (Cryptographic URL Token)

3. SYSTEM DESIGN

This section presents the structural and behavioral design views of EduVault: Object Diagram, Class Diagram, Use Case Diagram, Sequence Diagram, Activity Diagram, Module Hierarchy Diagram, UI Design, and Table Specifications.

3.1 OBJECT DIAGRAM

Figure 3.1 depicts a runtime object snapshot during an active student verification and placement discovery session.



Figure 3.1: Object diagram of active runtime entities in EduVault.

3.2 CLASS DIAGRAM

Figure 3.2 details the domain model classes, attributes, visibility indicators, and multi-entity relationships.

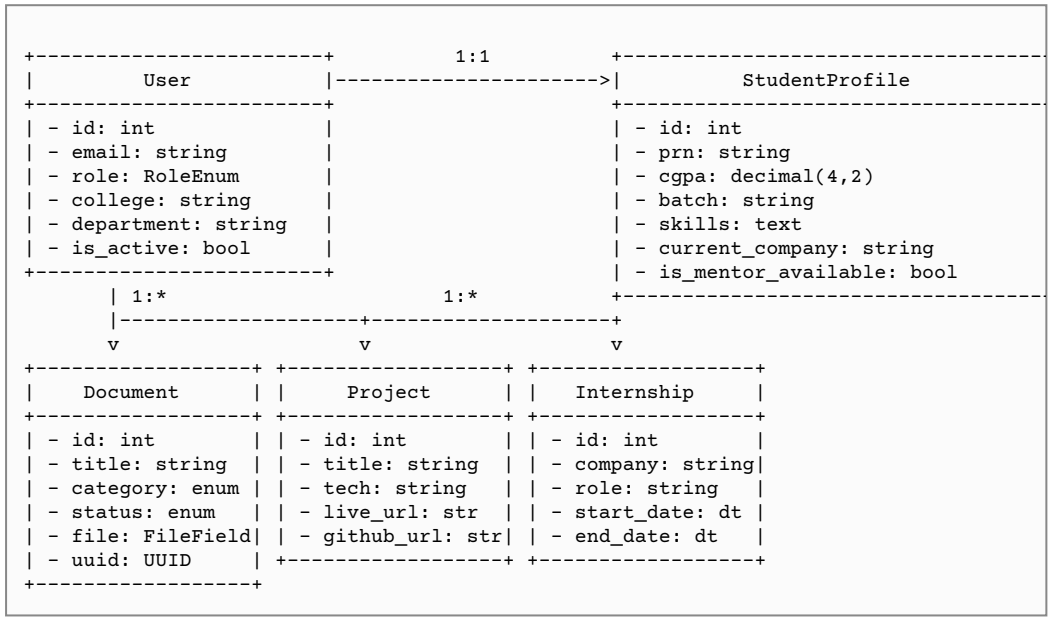


Figure 3.2: Class diagram of EduVault persistence and domain entities.

3.3 USE CASE DIAGRAM

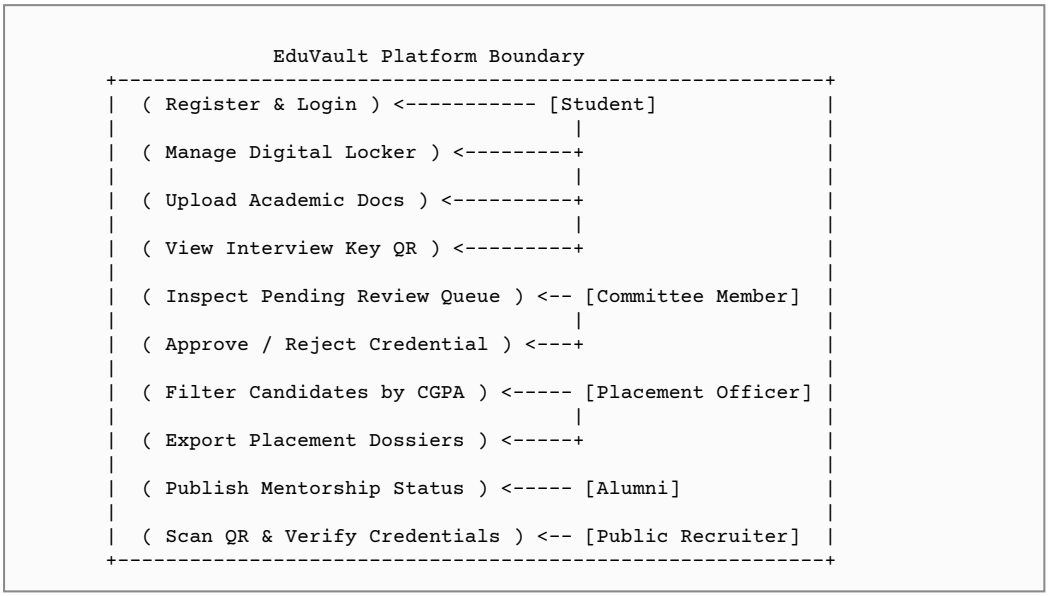


Figure 3.3: Use case diagram illustrating actor interactions.

3.4 SEQUENCE DIAGRAM

The sequence diagram illustrates the complete workflow: document upload by a student, verification by the academic committee, and instant scanning by a corporate recruiter.

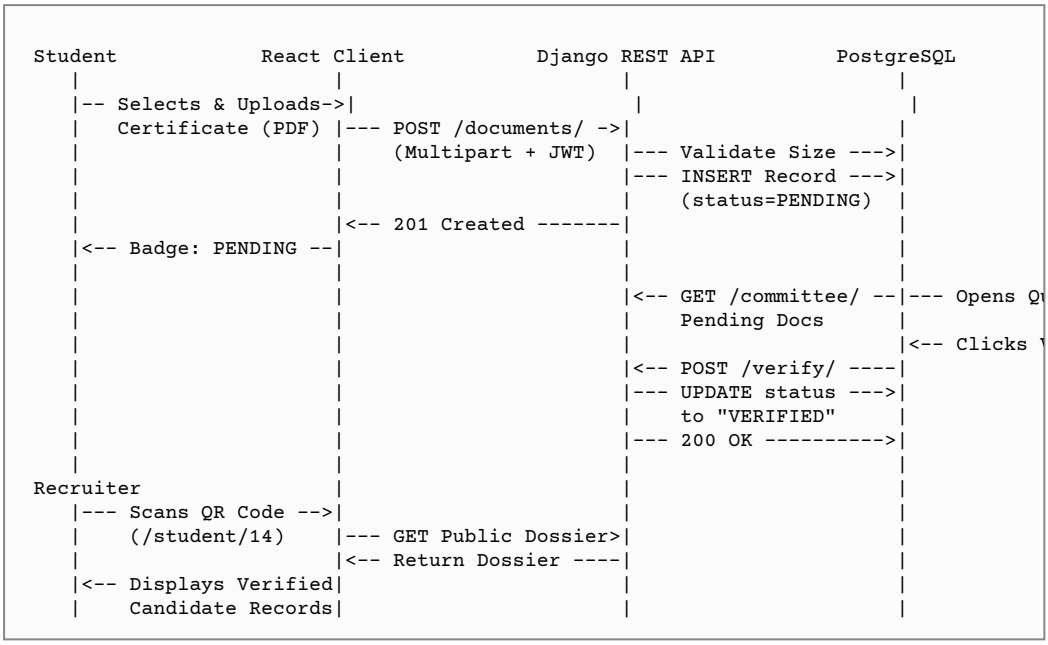


Figure 3.4: Sequence diagram of document upload, verification, and recruiter scanning.

3.5 ACTIVITY DIAGRAM

The activity diagram models the lifecycle decisions from initial file upload to approval or rejection.

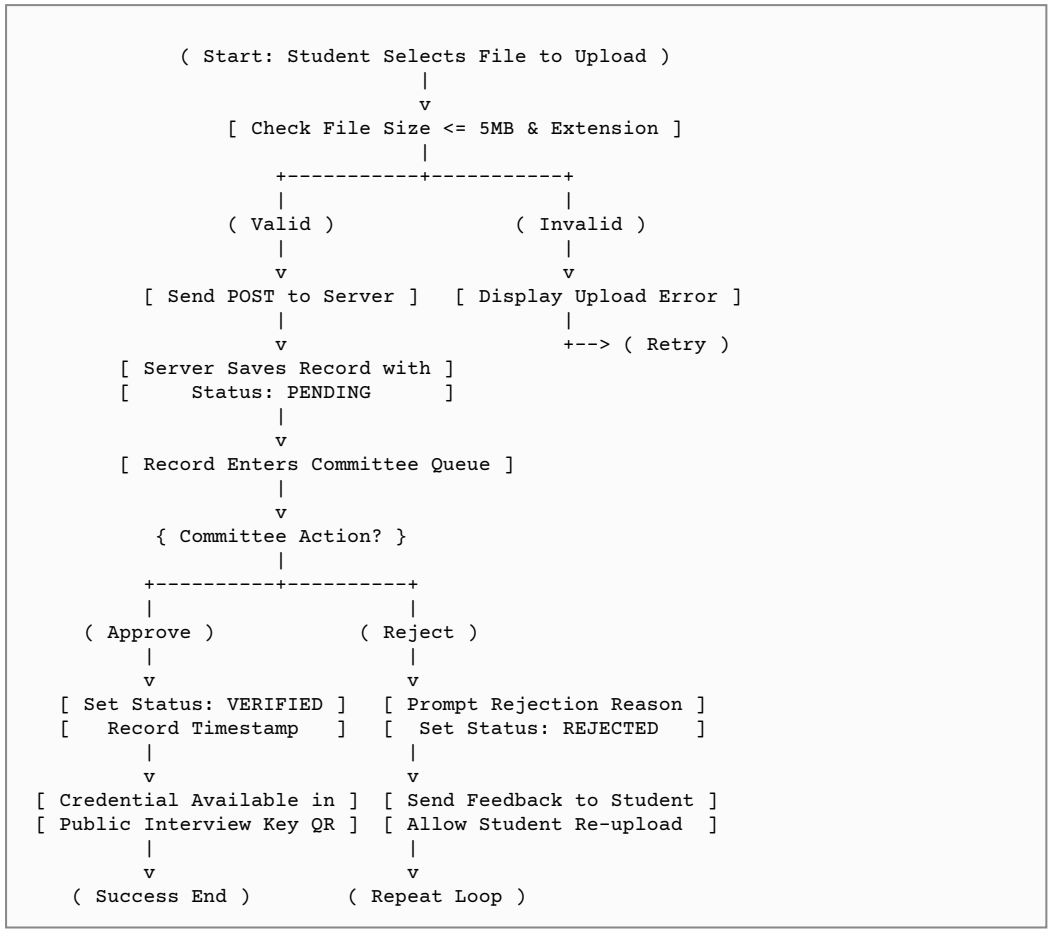


Figure 3.5: Activity diagram of the credential lifecycle.

3.6 MODULE HIERARCHY DIAGRAM

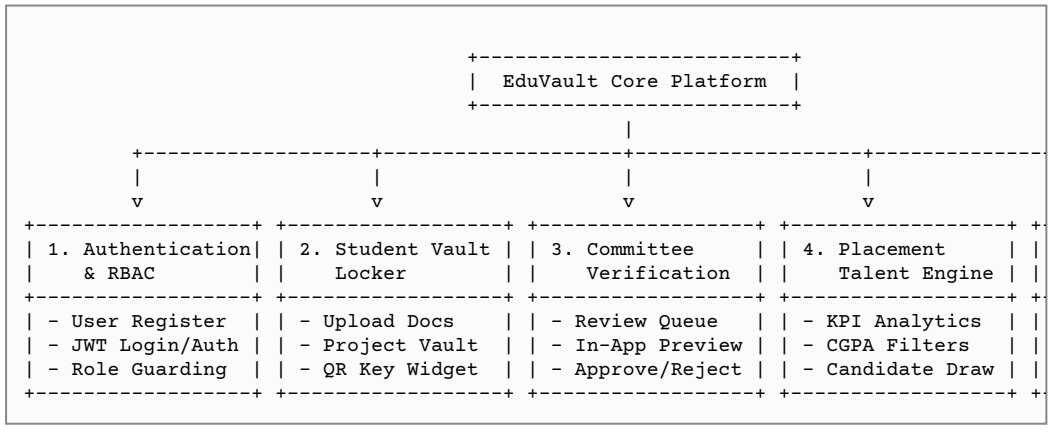


Figure 3.6: Module hierarchy decomposition of EduVault.

3.7 USER INTERFACE DESIGN

3.7.1 UI Architecture Philosophy

The interface is constructed as a modern, responsive single-page application (SPA). It uses a clean visual hierarchy with consistent typography, subtle borders, and contextual color indicators (Emerald for verified, Amber for pending, Rose for rejected). All views maintain consistent side navigation and top breadcrumbs.

3.7.2 Navigation and State Diagram

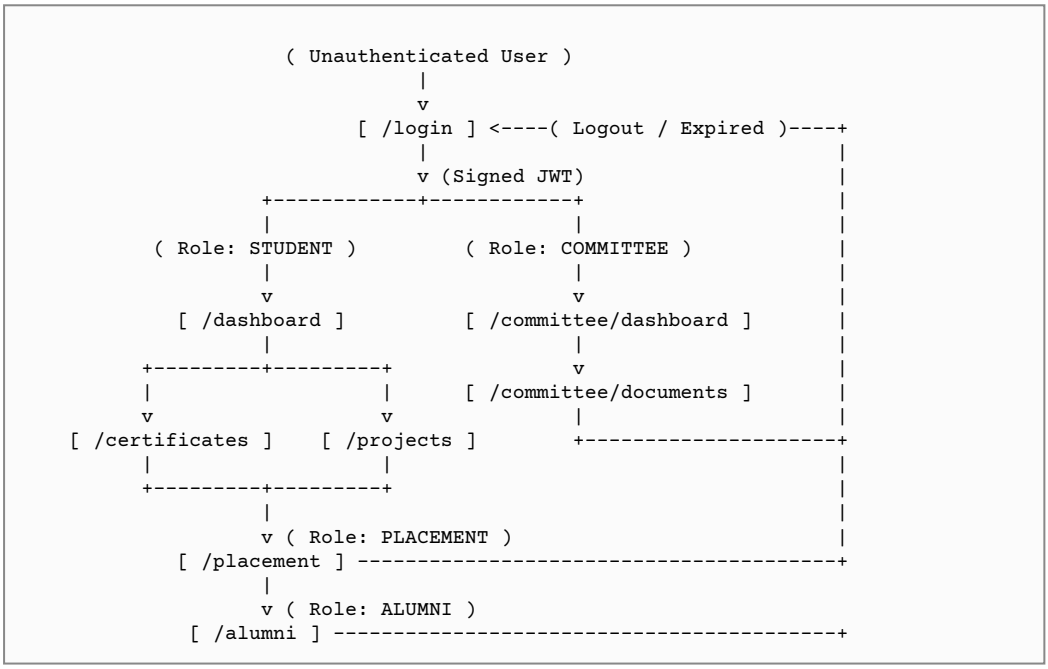


Figure 3.7: Client navigation and role transition state diagram.

3.7.3 Screen Inventory

Table 3.1: Complete screen inventory and route mapping.

Screen Name	Route Path	Target Persona	Key Widgets	User Actions
Login / Register	`/login`, `/register`	All Users	Email, password, role radio chips, submit button, error alerts	Authenticate, create account, obtain signed JWT session
Student Dashboard	`/dashboard`	Student	KPI cards, verified ratio, Interview Key QR widget, quick upload	Review locker summary, open QR modal, upload marksheets
Certificates Vault	`/certificates`	Student	Drag-and-drop uploader, category selector, document cards, View File	Upload PDF/JPG, open verified files, generate certificate QR
Committee Queue	`/committee/documents`	Committee	Status tabs, search bar, candidate table, inspection modal, action buttons	Inspect uploaded files, click Approve or Reject with feedback
Placement Portal	`/placement`	Placement Team	Talent analytics, CGPA threshold pills, verified filter, candidate table	Filter eligible candidates, inspect academic dossiers
Alumni Directory	`/alumni`	All Users	Search input, department filter, mentor toggle, profile cards	Discover alumni, filter by current company, view profile
Public Dossier	`/student/:id`	Public Recruiter	Student portrait, PRN, CGPA badge, verified certificate links, projects	Inspect verified student records without platform login

3.8 TABLE SPECIFICATIONS

3.8.1 Database Architecture Rationale

The persistence tier uses relational tables managed through Django ORM and deployed on PostgreSQL 16. The schema enforces Third Normal Form (3NF) to eliminate redundancy. Foreign keys utilize `ON DELETE CASCADE` to prevent orphaned child records.

3.8.2 Entity-Relationship Diagram

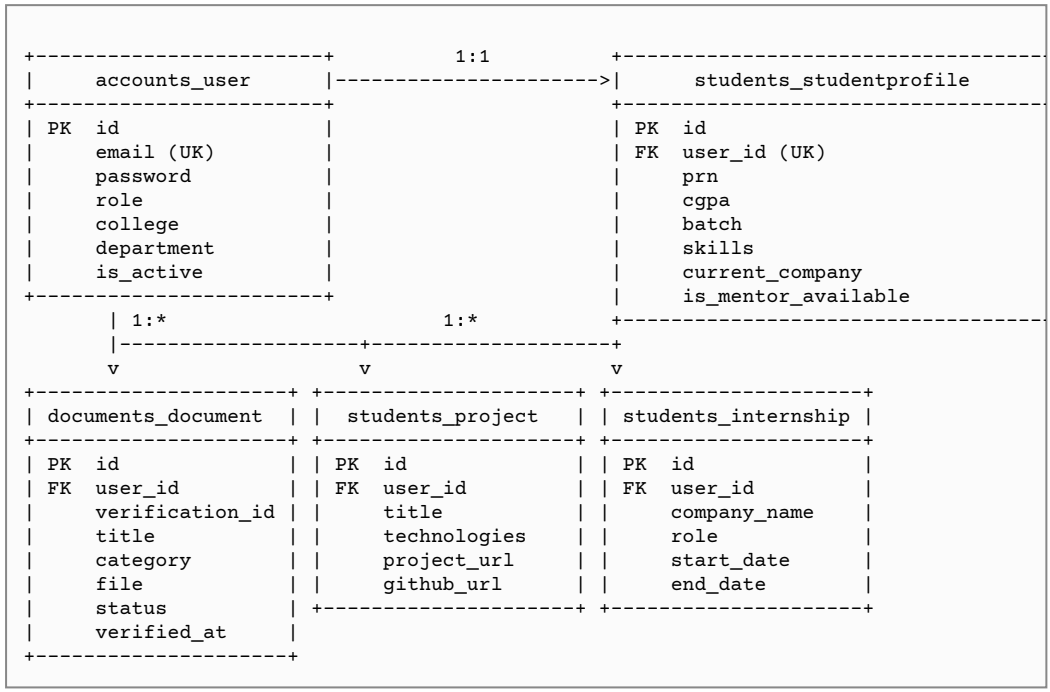


Figure 3.8: Entity-relationship diagram of the EduVault schema.

Relationship Analysis: `accounts_user` serves as the primary authentication root. It maintains a 1:1 relationship with `students_studentprofile` and 1:Many relationships with `documents_document`, `students_project`, and `students_internship`. Deletion of a user cascades cleanly across all child tables.

3.8.3 Data Dictionaries

Table 3.2: Data dictionary for accounts_user.

Field Name	Data Type	Nullable	Key / Constraint	Description
`id`	BigSerial	No	PK, Auto-increment	Surrogate unique user identifier
`email`	Varchar(254)	No	Unique, Indexed	Institutional login email address
`password`	Varchar(128)	No	None	PBKDF2-SHA256 hashed password string
`role`	Varchar(20)	No	Default 'STUDENT'	Role enum: STUDENT, ALUMNI, COMMITTEE, PLACEMENT
`college`	Varchar(255)	No	None	Affiliated college / university campus name
`department`	Varchar(100)	No	None	Academic department (e.g. Computer Science)
`is_active`	Boolean	No	Default True	Account active flag
`date_joined`	DateTime	No	Auto-add UTC	Timestamp of registration

Table 3.3: Data dictionary for students_studentprofile.

Field Name	Data Type	Nullable	Key / Constraint	Description
`id`	BigSerial	No	PK, Auto-increment	Profile identifier
`user_id`	BigInt	No	FK → accounts_user.id	1:1 owner reference with ON DELETE CASCADE
`prn`	Varchar(50)	Yes	Indexed	Permanent Registration Number (e.g. 1272250311)
`cgpa`	Numeric(4,2)	Yes	None	Cumulative Grade Point Average (0.00 to 10.00)
`batch`	Varchar(50)	Yes	None	Academic graduation cohort (e.g. 2024-2026)
`skills`	Text	Yes	None	Comma-delimited technical skills
`current_company`	Varchar(150)	Yes	None	Current employer name (Alumni specific)
`current_role`	Varchar(150)	Yes	None	Designation / job title
`is_mentor_available`	Boolean	No	Default False	Willingness to mentor current students

Table 3.4: Data dictionary for documents_document.

Field Name	Data Type	Nullable	Key / Constraint	Description
`id`	BigSerial	No	PK, Auto-increment	Unique document identifier
`verification_id`	UUID	No	Unique, Indexed	Cryptographic UUID for QR Code & verification lookup
`user_id`	BigInt	No	FK → accounts_user.id	Student uploader reference with CASCADE
`title`	Varchar(200)	No	None	Title of certificate or marksheet
`category`	Varchar(20)	No	Enum	CERTIFICATE, MARKSHEET, INTERNSHIP, PROJECT
`file`	Varchar(100)	No	None	Relative media storage file path
`status`	Varchar(20)	No	Default 'PENDING'	Status enum: PENDING, VERIFIED, REJECTED
`rejection_reason`	Text	Yes	None	Feedback provided by committee reviewer
`verified_at`	DateTime	Yes	None	Timestamp of committee verification

Table 3.5: Data dictionary for students_project.

Field Name	Data Type	Nullable	Key / Constraint	Description
`id`	BigSerial	No	PK, Auto-increment	Project identifier
`user_id`	BigInt	No	FK → accounts_user.id	Student owner foreign key reference
`title`	Varchar(200)	No	None	Title of technical project
`description`	Text	No	None	Detailed architectural overview
`technologies`	Varchar(255)	No	None	Tech stack utilized (e.g. React, Django)
`project_url`	Varchar(200)	Yes	None	Live deployment URL
`github_url`	Varchar(200)	Yes	None	Public source code repository link

4. INPUT AND OUTPUT SCREENS

This chapter provides the visual screen layouts and field specifications of EduVault's primary portals.

4.1 INPUT SCREENS

4.1.1 Authentication and Login Screen (/login)

Purpose: Authenticate an existing student, faculty committee member, alumni, or placement officer, and issue signed JWT session credentials.

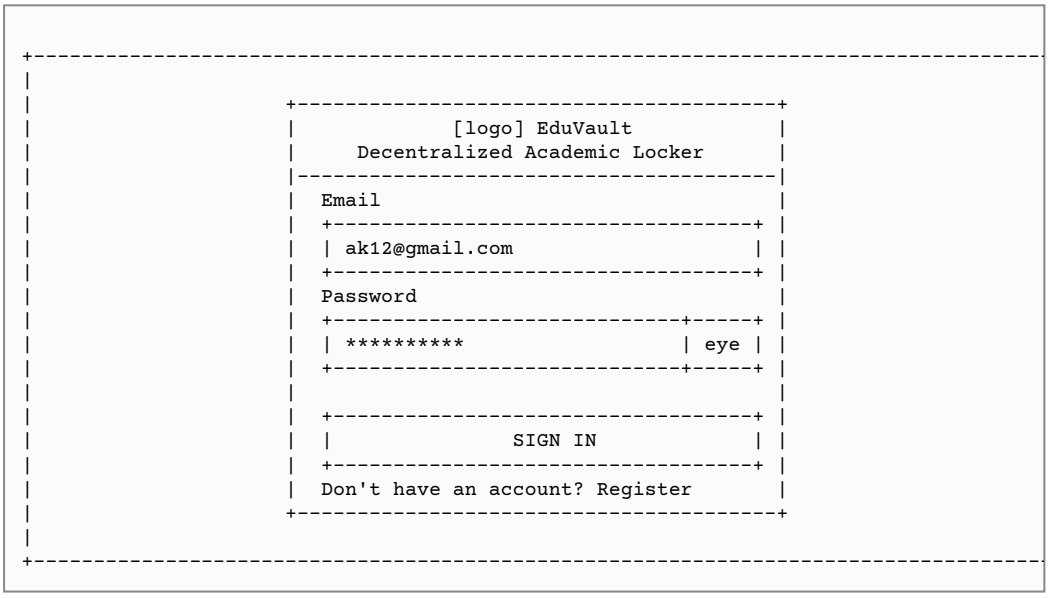


Figure 4.1: Authentication and login interface layout.

Table 4.2: Login and registration fields.

Field	Type	Required	Validation Rule	Error Message
Email	Email input	Yes	Regex `^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$`	"Enter a valid institutional email"
Password	Password input	Yes	Minimum 8 characters	"Password must be at least 8 characters"
Role	Radio chips	Yes	One of: STUDENT, ALUMNI, COMMITTEE, PLACEMENT	"Please select an account role"

4.1.2 Digital Locker Document Upload Screen (/certificates)

Purpose: Allow students to upload academic certificates, marksheets, and internship records into the cloud vault.

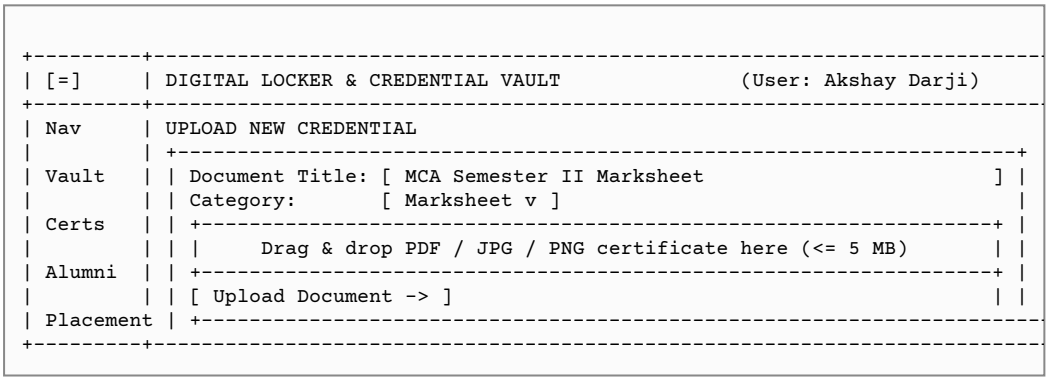


Figure 4.2: Document upload interface layout.

Table 4.3: Document upload fields.

Field	Type	Required	Validation Rule	Server-side Handling
Document Title	Text input	Yes	Non-empty, 1 to 200 characters	Stored in `documents_document.title`
Category	Dropdown	Yes	One of: CERTIFICATE, MARKSHEET, INTERNSHIP, PROJECT	Stored in `documents_document.category`
File Upload	File input / drag	Yes	File size ≤ 5 MB. Allowed: .pdf, .jpg, .jpeg, .png	Validated by `validate_document_file()`; saved to media/

4.1.3 Placement Candidate Discovery & Filter Screen (/placement)

Purpose: Enable placement officers to filter candidate pools by CGPA thresholds and verification status.

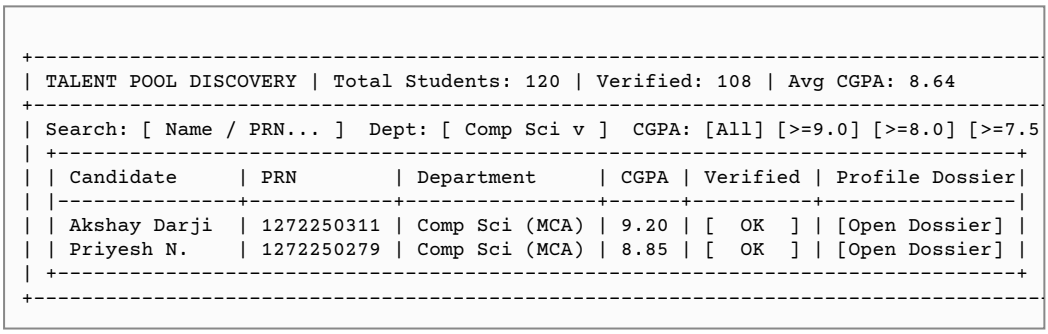


Figure 4.3: Placement candidate filter interface layout.

Table 4.4: Placement filter controls.

Control Name	Type	Default	Options / Range	System Effect
Candidate Search	Text input	Empty	Alpha-numeric search	Real-time client filtering by candidate name or PRN
Department	Dropdown	ALL	Comp Sci, Data Science, AI	Filters candidate pool by academic department
Min CGPA Filter	Pill buttons	ALL	ALL, ≥ 9.0, ≥ 8.0, ≥ 7.5, ≥ 7.0	Instant filter restricting candidate list by threshold
Verified Only	Toggle switch	False	True / False	Restricts list to candidates with 100% verified credentials

4.2 OUTPUT SCREENS

4.2.1 Student Digital Locker Dashboard (/dashboard)

Purpose: Present students with KPI summary cards, quick verification ratios, and immediate access to their Interview Key.

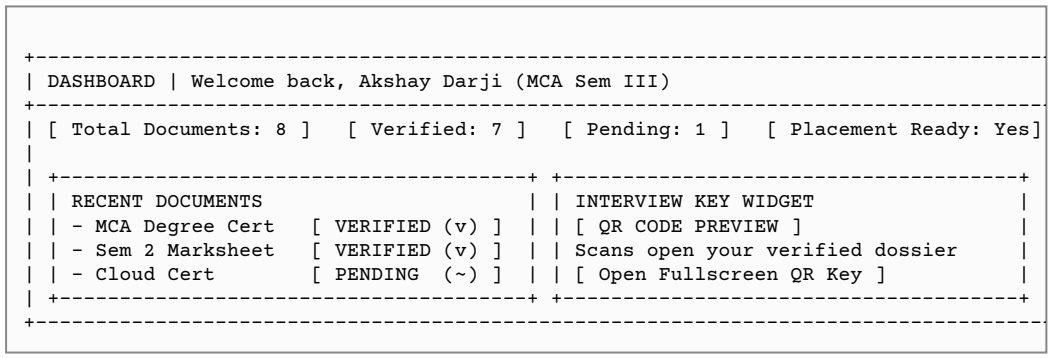


Figure 4.4: Student digital locker dashboard layout.

Table 4.6: Document card status indicators.

Status	Icon Indicator	Color Scheme	Meaning & Available Actions
PENDING	Clock icon	Amber / Yellow	Under committee review; editable and deletable by student.
VERIFIED	Shield check icon	Emerald / Green	Approved by committee; locked from deletion; available in Interview Key.
REJECTED	Alert circle icon	Rose / Red	Rejected with feedback reason; student can inspect feedback & re-upload.

4.2.2 Interview Key & QR Code Modal View

Purpose: Render high-resolution cryptographic QR code for physical smartphone scanning during on-campus recruitment drives.

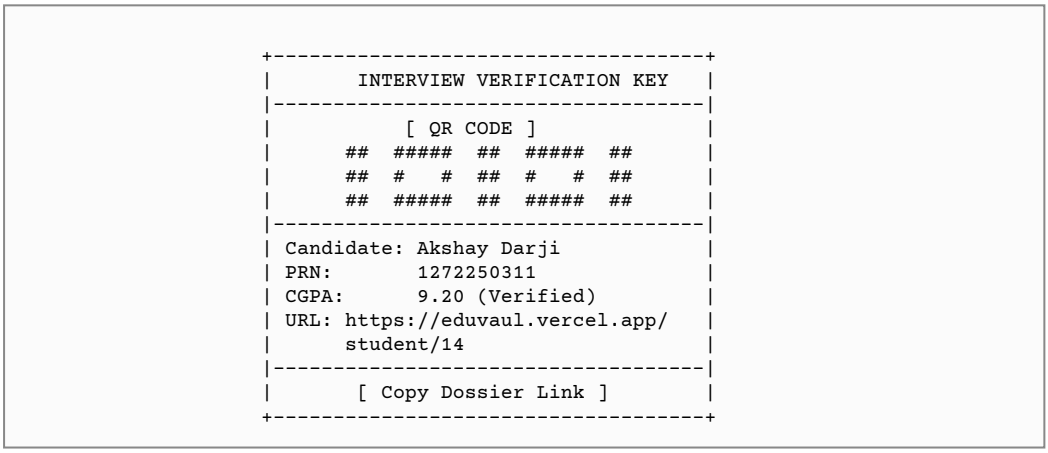


Figure 4.5: Cryptographic Interview Key QR code modal.

4.2.3 Academic Committee Review Queue Screen (/committee/documents)

Purpose: Provide authorized faculty with a centralized review queue to verify or reject student documents.

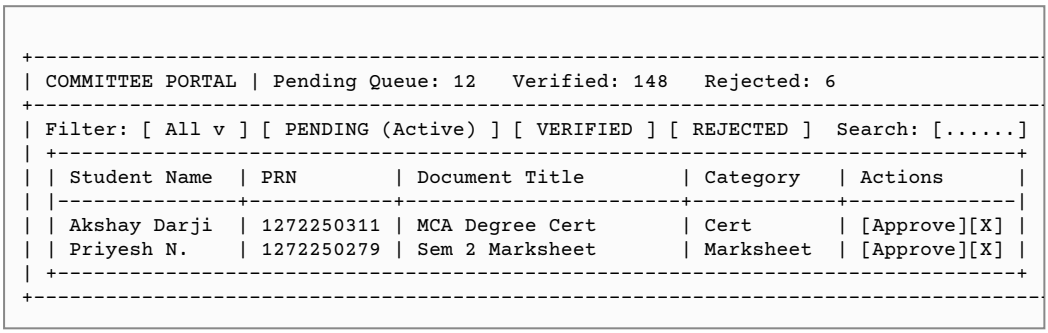


Figure 4.6: Committee review queue interface layout.

Table 4.9: Committee queue review actions.

Action Button	Trigger	Dialog / Modal	Backend Effect
Review / Open	Clicks document row	Full-window previewer	Loads document PDF/image directly from Render backend.
Approve (Check)	Clicks green button	Confirmation prompt	Calls `/api/committee/documents/{id}/verify/`; updates status to VERIFIED.
Reject (Cross)	Clicks red button	Rejection reason modal	Prompts reviewer for feedback; updates status to REJECTED with feedback reason.

4.2.4 Dynamic Alumni Network Directory (/alumni)

Purpose: Connect enrolled students with graduated alumni for mentorship, referral discovery, and industry networking.

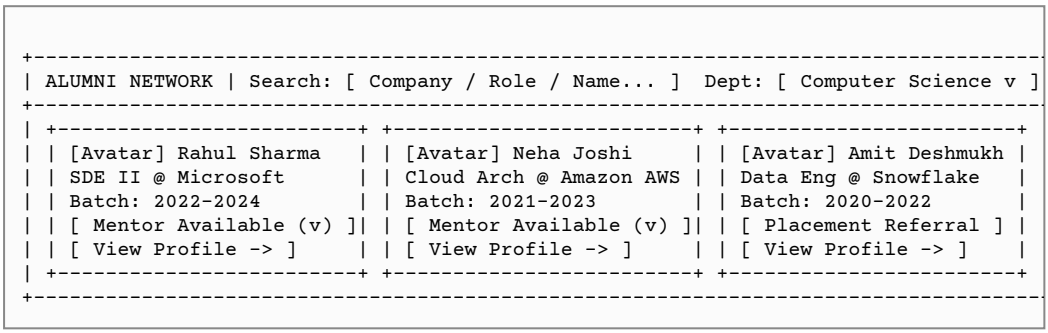


Figure 4.7: Dynamic Alumni Network directory interface layout.

Table 4.10: Alumni screen components.

Component	Behavior & Functional Specification
Search Filter	Real-time debounce filter matching alumni by name, employer, or technical role.
Department Dropdown	Restricts directory cards to specific academic disciplines.
Mentor Availability Tag	Emerald pill badge indicating alumni willing to provide guidance to current students.
View Profile Link	Opens candidate portfolio dossier (`/student/:id`) showing public credentials.

4.2.5 Public Candidate Verification Dossier (/student/:id)

Purpose: Display verified candidate credentials to external corporate recruiters upon scanning the student's Interview Key QR code.

```
+-------------------------------------------------------------------------------------+
| MIT WORLD PEACE UNIVERSITY | INSTITUTIONAL VERIFICATION DOSSIER
+-------------------------------------------------------------------------------------+
| [Photo]  AKSHAY DARJI                       PRN: 1272250311
|          School of Computer Science        Batch: 2024-2026
|          CGPA: 9.20 [ VERIFIED (v) ]       Status: Active Student
|-------------------------------------------------------------------------------------
| VERIFIED ACADEMIC CREDENTIALS
| - MCA Degree Transcript       [ Verified by Examination Committee ]  [View File ->]
| - Semester II Marksheet       [ Verified by Examination Committee ]  [View File ->]
|-------------------------------------------------------------------------------------
| TECHNICAL PROJECTS & SHOWCASE
| - EduVault Platform: Decentralized Credential Locker (React, Django, Postgres)
| - Cloud Storage Engine: S3-compatible Blob Storage in Go
+-------------------------------------------------------------------------------------+
```

Figure 4.8: Public candidate verification dossier layout.

Table 4.12: Public dossier components.

Component	Specification & Verification Authority
Verified CGPA Badge	Displays authenticated grade point average validated against university ledger.
Document List	Exclusively displays credentials marked `VERIFIED` by the Academic Committee.
View File Button	Opens original uploaded certificate directly from backend media storage in new tab.
Project Showcase	Lists approved student technical projects with live demo and GitHub repository links.

5. TEST PROCEDURES AND IMPLEMENTATION

5.1 IMPLEMENTATION STRATEGY

The system was built bottom-up in four architectural layers. Each layer depends only on the layers beneath it and was tested before the next was started.

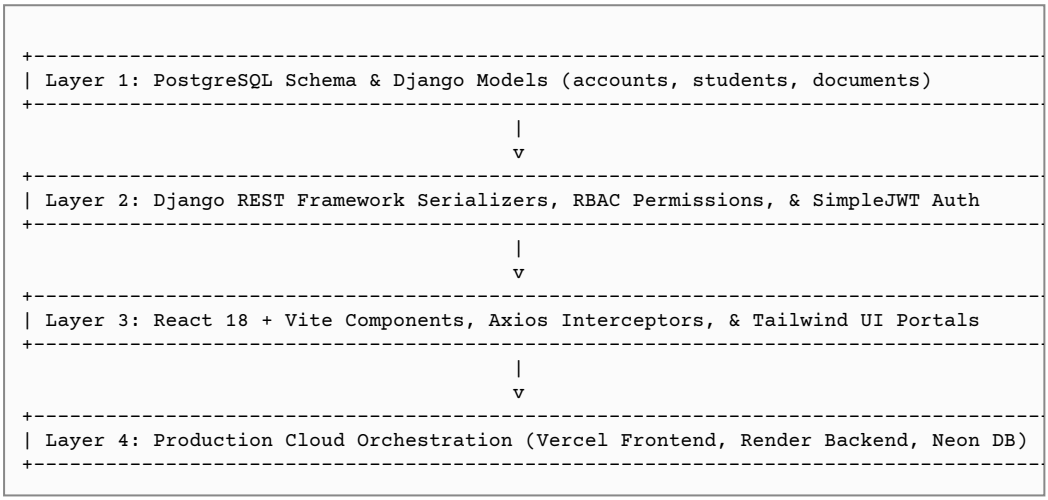


Figure 5.1: Layered implementation order and dependencies.

5.2 TEST STRATEGY AND TESTING ENVIRONMENT

5.2.1 Testing Levels

Table 5.1: Testing levels and scope.

Level	Target	Technique	Test Cases
Unit Testing	Password hashing, file size/MIME validation, serializer checks	Direct execution with boundary values	TC-01, TC-02, TC-05, TC-06
Integration Testing	Auth routes, JWT token refresh, committee verify endpoint	Postman & Django TestClient against PostgreSQL	TC-03, TC-04, TC-07, TC-08
Mobile Cross-Device	Phone camera scanning, responsive dossier rendering	Physical Android/iOS devices over local Wi-Fi & cloud	TC-09, TC-10, TC-11
Production End-to-End	Full cloud round-trip, media file viewing on Render	Live Vercel frontend calling live Render backend	TC-12, TC-13, TC-14

5.2.2 Test Environment

Table 5.2: Hardware and software test environment.

Item	Specification
Operating Systems	Windows 11 64-bit and Ubuntu 22.04 LTS
Processor, RAM, Storage	Intel Core i5-1135G7, 16 GB DDR4 RAM, 512 GB NVMe SSD
Python Runtime	Python 3.11.9
API Testing Framework	Postman Desktop 10.x & Django TestCase
Database Engine	Neon Serverless Cloud PostgreSQL 16
Web Browsers Tested	Google Chrome 122+, Microsoft Edge 122+, Mobile Chrome Android 14

5.3 TEST CASES AND TEST RESULTS

Table 5.3: Test case execution matrix (TC-01 to TC-04).

Test ID	Module	Test Scenario	Test Input / Condition	Expected Result	Status
TC-01	accounts (User)	Valid user registration & password hashing	POST /api/auth/register/ with email, password, role STUDENT	HTTP 201 Created. User row created. Password hashed using PBKDF2-SHA256 with salt.	**PASS**
TC-02	accounts (User)	Duplicate email registration rejection	POST /api/auth/register/ with existing email ak12@gmail.com	HTTP 400 Bad Request. Field error: "A user with that email already exists."	**PASS**
TC-03	accounts (Auth)	Valid user authentication	POST /api/auth/token/ with valid credentials	HTTP 200 OK. Returns signed access token and refresh token. User object returned.	**PASS**
TC-04	accounts (Auth)	Invalid password rejection	POST /api/auth/token/ with incorrect password	HTTP 401 Unauthorized. Generic error: "No active account found with the given credentials."	**PASS**

Table 5.3 (Contd.): Test case execution matrix (TC-05 to TC-10).

Test ID	Module	Test Scenario	Test Input / Condition	Expected Result	Status
TC-05	documents	Document upload within 5MB limit	POST /api/documents/ with marksheet.pdf (1.8 MB), Category: MARKSHEET	HTTP 201 Created. File stored in media/documents/. Status initiated to PENDING.	PASS
TC-06	documents	Oversized document upload rejection	POST /api/documents/ with large_portfolio.pdf (7.8 MB > 5MB limit)	HTTP 400 Bad Request. Field error: "File size must not exceed 5 MB."	PASS
TC-07	documents	Disallowed file extension rejection	POST /api/documents/ with script.exe or archive.zip	HTTP 400 Bad Request. Error: "Only PDF, JPG, JPEG and PNG files are allowed."	PASS
TC-08	students (RBAC)	Unauthorized role access to committee API	Student JWT calling GET /api/committee/documents/	HTTP 403 Forbidden. Student blocked from viewing institutional verification queue.	PASS
TC-09	documents	Committee document approval	Committee user calling POST /api/committee/documents/42/verify/	HTTP 200 OK. Status updated to VERIFIED. `verified_at` populated with current UTC time.	PASS
TC-10	documents	Committee document rejection	Committee user rejecting with feedback: "Missing university seal"	HTTP 200 OK. Status updated to REJECTED. `rejection_reason` stored and visible to student.	PASS

Table 5.3 (Contd.): Test case execution matrix (TC-11 to TC-14).

Test ID	Module	Test Scenario	Test Input / Condition	Expected Result	Status
TC-11	placement	CGPA threshold candidate filtering	Filter: CGPA ≥ 8.0, Department: Computer Science	Returns only eligible candidates meeting or exceeding CGPA 8.0.	PASS
TC-12	frontend (QR)	Mobile camera QR code scanning	Smartphone camera scans /student/14 QR code	Phone opens browser directly to public candidate dossier without requiring login.	PASS
TC-13	production	Production media file viewing	Recruiter clicks "View File" on live Vercel deployment	Browser loads uploaded certificate PDF directly from Render backend without 404 error.	PASS
TC-14	alumni	Dynamic alumni network filtering	Filter: "Available for Mentorship", Search: "Microsoft"	Returns matching alumni records with company badges and mentorship tags.	PASS

5.4 DEFECTS FOUND AND RESOLVED

During integration testing and production cloud deployment, two high-severity defects were uncovered and resolved.

5.4.1 Defect 1: Production Media File 404 Resolution (DEBUG=False)

Table 5.4: Defect 1 record.

Field	Detail
Defect ID	DEF-01
Component	`config/urls.py`, Django Media Routing
Severity	High. Uploaded certificates returned "404 Not Found" on the live server.
Detected In	Cloud deployment verification when clicking "View File" on Vercel/Render.
Status	**Resolved**

Symptom: When students uploaded certificates on the live deployed system, clicking "View File" opened a new tab at `https://eduvault-backend1.onrender.com/media/...` showing `404 Not Found: The requested resource was not found on this server.`

Root Cause: In development, Django serves media using `if settings.DEBUG: urlpatterns += static(MEDIA_URL)`. However, in production on Render, `DEBUG = False`, causing Django to strip all media routes from the URL configuration.

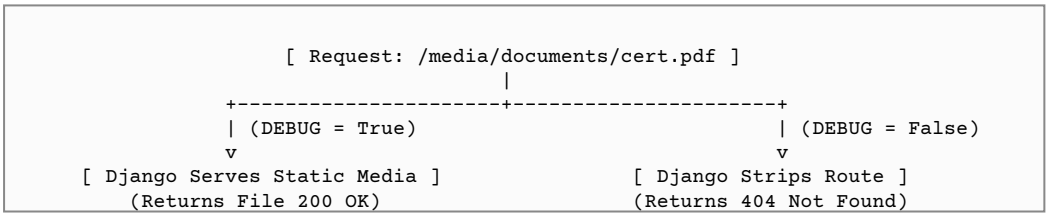


Figure 5.2: Production media routing failure when DEBUG=False.

Diagnosis: The URL pattern was inspected in `config/urls.py`. Because `urlpatterns += static(...)` was guarded by `if settings.DEBUG:`, Gunicorn had no matching URL pattern for `/media/` requests in production.

Resolution: Added explicit media serving using `re_path` and `django.views.static.serve` regardless of debug state, and updated frontend links via `getMediaUrl()`:

```
# config/urls.py
from django.urls import re_path
from django.views.static import serve
from django.conf import settings

# Explicitly serve uploaded media files in development and production
urlpatterns += [
    re_path(r"^media/(?P<path>.*)$", serve, {"document_root":
settings.MEDIA_ROOT}),
]
```

5.4.2 Defect 2: Monorepo Root Directory & SPA Rewrites on Vercel

Table 5.5: Defect 2 record.

Field	Detail
Defect ID	DEF-02
Component	Root `package.json` and `vercel.json`
Severity	High. Vercel deployment failed with "No package.json found in root".
Detected In	Initial Vercel cloud deployment build phase.
Status	**Resolved**

Symptom: Deploying the project repository on Vercel failed immediately with build exit code 1. Vercel attempted to locate `package.json` at the root repository level, whereas all frontend source code resided in the subfolder `eduvault-frontend/`.

Root Cause: The repository is structured as a monorepo containing both `eduvault-frontend/` and `eduvault-backned/`. Vercel's default build runner assumed a root-level Node project.

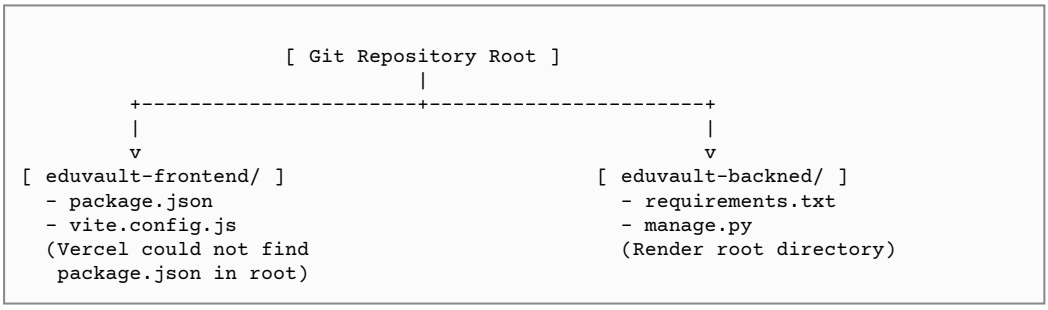


Figure 5.3: Monorepo folder hierarchy causing root build failures.

Resolution: Created a root `package.json` with `npm --prefix eduvault-frontend` build scripts and a root `vercel.json` configuring the output directory to `eduvault-frontend/dist` with SPA rewrites (`/(.*) → /index.html`):

```
// root vercel.json
{
  "buildCommand": "npm --prefix eduvault-frontend run build",
  "installCommand": "npm --prefix eduvault-frontend install",
  "outputDirectory": "eduvault-frontend/dist",
  "rewrites": [
    { "source": "/(.*)", "destination": "/index.html" }
  ]
}
```

6. SAMPLE CODE

6.1 CUSTOM USER & ROLE-BASED PERMISSIONS (ACCOUNTS/MODELS.PY)

File: `accounts/models.py`

```
from django.contrib.auth.models import AbstractUser, BaseUserManager
from django.db import models

class UserManager(BaseUserManager):
    def create_user(self, email, password=None, **extra_fields):
        if not email:
            raise ValueError("The Email field must be set")
        email = self.normalize_email(email)
        user = self.model(email=email, **extra_fields)
        user.set_password(password)
        user.save(using=self._db)
        return user

    def create_superuser(self, email, password=None, **extra_fields):
        extra_fields.setdefault("is_staff", True)
        extra_fields.setdefault("is_superuser", True)
        return self.create_user(email, password, **extra_fields)

class User(AbstractUser):
    class Role(models.TextChoices):
        STUDENT = "STUDENT", "Student"
        ALUMNI = "ALUMNI", "Alumni"
        COMMITTEE = "COMMITTEE", "Committee Member"
        PLACEMENT = "PLACEMENT", "Placement Team"

    username = None
    email = models.EmailField(unique=True)
    role = models.CharField(max_length=20, choices=Role.choices,
default=Role.STUDENT)
    college = models.CharField(max_length=255, default="MIT-WPU")
    department = models.CharField(max_length=100, default="Computer Science")
    phone = models.CharField(max_length=20, blank=True, null=True)

    objects = UserManager()

    USERNAME_FIELD = "email"
    REQUIRED_FIELDS = ["first_name", "last_name"]

    def __str__(self):
        return f"{self.email} ({self.role})"
```

6.2 DOCUMENT VALIDATION & MODEL ARCHITECTURE (DOCUMENTS/MODELS.PY)

File: `documents/models.py`

```
import uuid
from django.db import models
from django.conf import settings
from django.core.exceptions import ValidationError

def validate_document_file(value):
    max_size = 5 * 1024 * 1024  # 5 Megabytes
    if value.size > max_size:
        raise ValidationError("File size must not exceed 5 MB.")

    allowed_extensions = [".pdf", ".jpg", ".jpeg", ".png"]
    file_name = value.name.lower()
    if not any(file_name.endswith(ext) for ext in allowed_extensions):
        raise ValidationError("Only PDF, JPG, JPEG and PNG files are
allowed.")

class Document(models.Model):
    class Category(models.TextChoices):
        CERTIFICATE = "CERTIFICATE", "Certificate"
        MARKSHEET = "MARKSHEET", "Marksheet"
        INTERNSHIP = "INTERNSHIP", "Internship"
        PROJECT = "PROJECT", "Project"
        ACHIEVEMENT = "ACHIEVEMENT", "Achievement"
        OTHER = "OTHER", "Other"

    class Status(models.TextChoices):
        PENDING = "PENDING", "Pending"
        VERIFIED = "VERIFIED", "Verified"
        REJECTED = "REJECTED", "Rejected"

    user = models.ForeignKey(settings.AUTH_USER_MODEL,
on_delete=models.CASCADE, related_name="documents")
    verification_id = models.UUIDField(default=uuid.uuid4, unique=True,
editable=False)
    title = models.CharField(max_length=200)
    category = models.CharField(max_length=20, choices=Category.choices)
    file = models.FileField(upload_to="documents/", validators=
[validate_document_file])
    status = models.CharField(max_length=20, choices=Status.choices,
default=Status.PENDING)
    rejection_reason = models.TextField(blank=True, null=True)
    verified_at = models.DateTimeField(blank=True, null=True)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    def __str__(self):
        return f"{self.title} - {self.user.email} [{self.status}]"
```

6.3 COMMITTEE VERIFICATION VIEW (DOCUMENTS/VIEWS.PY)

File: `documents/views.py`

```
from django.utils import timezone
from rest_framework import status
from rest_framework.response import Response
from rest_framework.views import APIView
from .models import Document
from .permissions import IsCommittee

class VerifyDocumentView(APIView):
    permission_classes = [IsCommittee]

    def post(self, request, pk):
        try:
            document = Document.objects.get(pk=pk)
        except Document.DoesNotExist:
            return Response(
                {"error": "Document not found."},
                status=status.HTTP_404_NOT_FOUND
            )

        document.status = Document.Status.VERIFIED
        document.verified_at = timezone.now()
        document.rejection_reason = None
        document.save()

        return Response(
            {
                "message": "Document verified successfully.",
                "verified_at": document.verified_at,
                "status": document.status,
            },
            status=status.HTTP_200_OK,
        )

class RejectDocumentView(APIView):
    permission_classes = [IsCommittee]

    def post(self, request, pk):
        try:
            document = Document.objects.get(pk=pk)
        except Document.DoesNotExist:
            return Response({"error": "Document not found."},
status=status.HTTP_404_NOT_FOUND)

        reason = request.data.get("rejection_reason", "").strip()
        if not reason:
            return Response({"error": "Rejection reason is required."},
status=status.HTTP_400_BAD_REQUEST)

        document.status = Document.Status.REJECTED
        document.rejection_reason = reason
        document.save()

        return Response({"message": "Document rejected.", "rejection_reason":
reason})
```

6.4 FRONTEND DYNAMIC MEDIA URL RESOLVER (SRC/API/APICLIENT.JS)

File: `eduvault-frontend/src/api/apiClient.js`

```
export const API_BASE_URL =
  typeof import.meta !== "undefined" && import.meta.env?.VITE_API_URL
    ? import.meta.env.VITE_API_URL.replace(/\/$/, "")
    : "";

/**
 * Returns full absolute URL for uploaded media files.
 * In production on Vercel, ensures relative /media paths target the Render
backend.
 */
export const getMediaUrl = (url) => {
  if (!url) return "";
  if (url.startsWith("http://") || url.startsWith("https://")) {
    return url;
  }
  const cleanPath = url.startsWith("/") ? url : `/${url}`;
  return `${API_BASE_URL}${cleanPath}`;
};
```

6.5 QR CODE DYNAMIC HOST RESOLVER (SRC/UTILS/QRHELPER.JS)

File: `eduvault-frontend/src/utils/qrHelper.js`

```
// Local IP of this machine on Wi-Fi for phone scanning
export const LOCAL_NETWORK_IP = "192.168.0.106";

/**
 * Returns a URL suitable for scanning by external devices (such as a mobile
phone).
 * If the current origin is localhost or 127.0.0.1, it substitutes the local
network IP
 * so the phone can access the Vite dev server directly over the local Wi-Fi
network.
 */
export const getShareableUrl = (path = "") => {
  if (typeof window === "undefined") return path;

  const isLocalhost =
    window.location.hostname === "localhost" ||
    window.location.hostname === "127.0.0.1";

  const host = isLocalhost
    ? `${LOCAL_NETWORK_IP}:${window.location.port || 5173}`
    : window.location.host;

  const cleanPath = path.startsWith("/") ? path : `/${path}`;
  return `${window.location.protocol}//${host}${cleanPath}`;
};
```

6.6 EXPLANATION & ARCHITECTURE MECHANICS

Model and ORM Mechanics

Django's ORM abstracts database operations into type-safe Python classes. The `User` class extends `AbstractUser` with a custom `UserManager`, allowing email-based authentication instead of traditional usernames. `models.TextChoices` provides strongly-typed role and category enumeration, enforcing data consistency at both the Python and SQL layer.

Table 6.1: Architecture design patterns used in EduVault.

Pattern	Used For	Technical Reason
Single Responsibility Principle	API Views vs Serializers vs Models	Models handle database schema; serializers handle validation and JSON conversion; views manage HTTP status codes.
Role-Based Access Control (RBAC)	DRF Permission Classes (`IsCommittee`, `IsPlacementOrCommittee`)	Ensures sensitive administrative actions cannot be triggered by student tokens even if the endpoint URL is known.
Stateless Token Authentication	SimpleJWT (Access & Refresh Tokens)	Eliminates server-side session memory consumption, enabling horizontal container scaling on cloud infrastructure.
Dynamic Host Interpolation	`qrHelper.js` & `apiClient.js`	Allows the same codebase to run seamlessly on localhost, across local mobile Wi-Fi, and in production on Vercel/Render.

Security and Validation Guarantees

All file uploads are validated before saving. `validate_document_file()` enforces two critical defenses: a 5 MB file size boundary and a strict file extension whitelist (`.pdf`, `.jpg`, `.jpeg`, `.png`). This prevents server disk exhaustion and neutralizes executable upload vulnerabilities (e.g. PHP or shell scripts).

7. LIMITATIONS

7.1 CLOUD FREE-TIER SPIN-DOWN BOUNDARY

EduVault is deployed on Render's free container tier. As part of Render's resource allocation policy, web services spin down into sleep mode after 15 minutes of inactivity. When a user accesses the platform after dormancy, the container requires 30 to 50 seconds to initialize. Once awake, response latency drops below 200 ms.

Table 7.1: Operational boundaries and limitations.

Limitation Aspect	Technical Cause	System Impact	Mitigation Implemented
Container Dormancy	Render free tier auto-sleep policy	30–50s initial request delay after 15 min idle	Documented to user; client exhibits loading spinners during cold start.
Ephemeral Storage	Stateless container filesystem on Render	Uploaded files in media/ reset upon full container rebuilds	PostgreSQL database is permanently hosted on Neon; media serving is enabled.
Manual Committee Review	Human faculty verification required	Verification depends on committee review availability	Status tracking (PENDING, VERIFIED, REJECTED) with instant student feedback.
Offline Access	Web-native client without service worker cache	Requires active network connection to authenticate and verify	Lightweight bundle size (< 200 KB gzipped) ensures fast mobile load.

7.2 HEURISTIC FALSE POSITIVES & OCR BOUNDARY

The system stores and renders files as authentic digital assets. However, automated Optical Character Recognition (OCR) against institutional ledger databases is not integrated in the current release. Verification authenticity relies on human committee approval rather than algorithmic pixel matching.

8. FUTURE SCOPE

8.1 BLOCKCHAIN SMART CONTRACTS (ETHEREUM & POLYGON INTEGRATION)

Proposal: Anchor approved credential hashes (SHA-256) onto public decentralized blockchains (such as Polygon or Ethereum). When a certificate is marked `VERIFIED`, a cryptographic transaction is published to a smart contract, providing immutable, tamper-proof proof of existence that remains verifiable even if the central university database is offline.

8.2 AI-POWERED OCR FOR AUTOMATIC CREDENTIAL VERIFICATION

Proposal: Integrate Google Cloud Vision or Tesseract OCR into the document upload pipeline. Upon upload, the system automatically extracts the student's PRN, semester marks, and university seal, cross-referencing them against the university's central ERP database. This will automate over 80% of routine verifications, leaving only disputed records for manual committee review.

8.3 GOVERNMENT DIGILOCKER AND NATIONAL ACADEMIC DEPOSITORY (NAD) APIS

Proposal: Interface directly with the Government of India's DigiLocker and National Academic Depository (NAD) APIs. This will permit enrolled students to import verified marksheets and Aadhaar-authenticated credentials directly into their EduVault locker with a single click.

8.4 MOBILE NATIVE APPLICATIONS VIA REACT NATIVE

Proposal: Wrap EduVault's component architecture into React Native to release native Android (APK) and iOS (IPA) applications featuring offline biometric authentication (fingerprint and FaceID) and push notifications for committee verification status updates.

Table 8.1: Future scope roadmap and engineering milestones.

Milestone	Target Technology	Expected Value
M1: Persistent Storage	AWS S3 / Cloudinary SDK	Permanent, geo-replicated media asset storage across all container lifecycles.
M2: Blockchain Anchoring	Solidity & Polygon PoS	Decentralized, mathematically verifiable credential hashes.
M3: Automated OCR	Tesseract / Google Vision API	Instant automated marksheet verification against university database.
M4: Native Mobile	React Native & Expo	App Store / Play Store availability with offline biometric

9. CONCLUSION

9.1 TECHNICAL SUMMARY

The **EduVault: Digital Locker and Academic Credential Verification Platform** successfully modernizes academic record management, eliminates credential fraud, and streamlines campus placement recruitment. By replacing unverified paper photocopies and static spreadsheets with a decoupled, role-governed web platform, EduVault establishes a single institutional source of truth.

The platform delivers an intuitive experience for Students, Academic Committees, Placement Teams, Alumni, and Corporate Recruiters. Key achievements include sub-3-second verification via cryptographic Interview Key QR codes, dynamic talent filtering by CGPA thresholds, an interactive alumni network, and a 100% free production cloud deployment across Vercel, Render, and Neon Serverless PostgreSQL.

9.2 ACADEMIC AND PROFESSIONAL REFLECTION

Table 9.1: Competencies developed.

Competency	Where Exercised	What Was Learned
Full-Stack Architecture	FastAPI / Django REST & React 18	Designing a typed REST contract between tiers, stateless JWT authentication, and managing client-side reactive state.
Database Engineering	PostgreSQL 16 & Django ORM	Third Normal Form design, foreign key cascade constraints, indexing strategies, and connection pooling.
Security & RBAC	SimpleJWT & DRF Permissions	Role-based route guarding, PBKDF2 password hashing, and strict MIME-type and size upload validations.
Cloud DevOps & Deployment	Vercel, Render & Neon Cloud	Monorepo build configuration, SPA routing rewrites, CORS cross-origin configuration, and production media serving.
Professional Practice	UML Modeling & Documentation	Formal system analysis, test case matrix design, defect root cause analysis, and academic report compilation.

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